DS5007 Assignment 2: Neural NLP Pipeline for BBC Urdu

i24-7709 (DS-1A) | Repository: https://github.com/SarmadRazaq/i24-7709-NLP-Assignment2

1. Overview

This assignment extends the Assignment 1 pipeline (300 BBC Urdu articles, 460k tokens) in three stages. Part 1 builds count-based (TF-IDF, PPMI) and neural (Skip-gram) word embeddings. Part 2 uses them in BiLSTM POS and NER taggers, the NER tagger with a CRF layer. Part 3 trains a Transformer encoder for 5-class topic classification. Everything is implemented from scratch in PyTorch and NumPy, and no pretrained models or built-in Transformer modules are used. Metadata.json has no section labels, so articles were assigned to the five required categories by keyword scoring (Politics 48, Sports 94, Economy 12, International 55, Health & Society 91). POS and NER gold labels are silver annotations produced by a rule-based tagger. It uses a hand-crafted lexicon (NOUN 288, VERB 330, ADJ 235, ADV 266 entries) and a gazetteer (111 Pakistani persons, 124 locations, 85 organisations). The lexicon is built on the Assignment 1 stemmer.

2. Part 1 Results: Word Embeddings

TF-IDF (300 × 10002) gives clearly topical discriminative words, e.g. Sports: cricket, match, runs, player, wicket; Politics: police, court, assembly, minister. PPMI (window 5) neighbours are strong for collocated words (court → court/supreme/judge/hearing) but noisy for general nouns. In the t-SNE map of the 200 most frequent words, function words, auxiliaries, numbers and sports terms form separate clusters. The PPMI matrix is stored for the 5,000 most frequent words because of GitHub's file-size limit. Skip-gram (V and U matrices, K = 10, unigram^0.75 noise, Adam 1e-3, batch 1024, 15 epochs, subsampling 1e-4) reduced the loss from about 3.3 to 2.51. C3 solves 8/10 analogies (e.g. Pakistan:Lahore :: India:Delhi; Lionel:Messi :: Cristiano:Ronaldo; man:woman :: boy:girl). Capital-city and some kinship analogies fail.

Condition	MRR (20 pairs)	Analogies
C1 PPMI	0.431	4/10
C2 SG raw	0.237	4/10
C3 SG cleaned	0.229	8/10
C4 SG d=200	0.263	7/10

PPMI scores the highest MRR because the gold pairs are mostly direct collocates, which PPMI measures by construction. C3 gives the most coherent neighbours and the best analogy performance, so it is used in Part 2. Raw text (C2) splits counts across inflected surface forms. Doubling d (C4) gives only a small MRR gain and fewer correct analogies: 460k tokens are too few to exploit 200

dimensions. Two preprocessing artefacts are visible. The repeated BBC footer pulls 'Pakistan' toward boilerplate words, and the stemmer merges 'aabadi' (population) with '-abad' city names.

3. Part 2 Results: POS Tagging and NER

500 sentences (125 per topic from 4 topics) were split 70/15/15 stratified by topic (7757/1838/1658 tokens). The 2-layer BiLSTM (128 units per direction, dropout 0.5) was initialised with C3 embeddings and trained with Adam (1e-3, wd 1e-4) and early stopping on validation F1. NER uses a CRF with Viterbi decoding.

Task	Embeddings	Val F1	Test acc (POS) / P (NER)	Test macro-F1 / entity F1
POS	frozen	65.1	80.5	68.3
POS	fine-tuned	84.2	94.3	92.1
NER	frozen	52.4	50.0	47.2
NER	fine-tuned	69.3	74.3	63.4

POS reaches 94.3% token accuracy. The most confused pairs are NOUN <-> ADJ (45), NOUN <-> ADV (14) and NOUN <-> VERB (8). Stemming removes the inflections that separate adjectives from nouns, and the rule-based labels for unknown content words are themselves noisy. With conlleval, NER scores LOC F1 87.0, PER 24.0, ORG 66.7, MISC 50.0 (overall P 74.3, R 55.3, F1 63.4). Locations are learned well because they recur, while person names are rare and mostly unseen. Several false positives are in fact better than the silver labels (e.g. a full three-token name where the rules marked two tokens).

Task	Setting (3 seeds)	Test macro-F1 / entity F1
POS	Full model (fine-tuned)	92.9 ± 0.6
POS	A1 unidirectional LSTM	88.9 ± 1.1
POS	A2 no dropout	93.1 ± 0.4
POS	A3 random embedding init	86.8 ± 0.6
NER	Full model (fine-tuned)	63.4 ± 1.4
NER	A1 unidirectional LSTM	53.6 ± 3.9
NER	A2 no dropout	64.3 ± 1.7
NER	A3 random embedding init	58.1 ± 1.7
NER	A4 softmax instead of CRF	62.0 ± 3.8

Backward context matters most for NER (A1: -9.8 F1 points), because Urdu postpositions and titles that mark an entity occur on both sides of it. Pre-trained embeddings help both tasks (A3: POS -6.1, NER -5.3). Removing dropout (A2) has no significant effect with early stopping on this small dataset. The CRF (A4) gives a small, consistent gain over softmax (1.4 points) with lower variance, because it prevents invalid transitions such as O → I-PER.

4. Part 3 Results: Transformer Topic Classifier

The Transformer (4 Pre-LN blocks, 4 heads, d_model 128, d_ff 512, sinusoidal PE, [CLS] + MLP; 2,082,309 parameters) was trained with AdamW (5e-4, wd 0.01) and a cosine schedule with 50 warm-up steps for 20 epochs on 210 articles truncated to 256 tokens. It reaches test accuracy 64.4% and macro-F1 50.8. The macro-F1 is pulled down by the tiny Economy class (2 test articles). A BiLSTM classifier baseline on the same split gives accuracy 66.7% and macro-F1 50.4. The Transformer reached its best validation accuracy at epoch 10, the BiLSTM at epoch 7. On CPU, one epoch took 8.75 s for the Transformer and 1.89 s for the BiLSTM. Final-layer attention heatmaps show [CLS] concentrating on topical content words (e.g. cricket/test/player for a sports article), and heads specialise: some attend to a few keywords, others spread almost uniformly. With only 200–300 articles both models overfit: training accuracy approaches 97% while validation loss rises after epochs 7–13. The 2.2-point accuracy gap equals one test article. The BiLSTM with pre-trained embeddings is the more appropriate choice at this scale. It has a stronger inductive bias and starts from distributional knowledge, while the Transformer has to learn its embeddings and attention patterns from scratch.

5. Conclusion

On a small, heavily preprocessed Urdu corpus, simple methods remain competitive. PPMI is a strong baseline for direct relatedness, while Skip-gram embeddings capture more transferable semantics and clearly help downstream tagging, especially when fine-tuned. Bidirectional context and pre-trained initialisation were the most important BiLSTM components, and the CRF gave a smaller but consistent gain for NER. The from-scratch Transformer works and produces interpretable attention, but with roughly 200 training documents it does not beat a recurrent baseline with pre-trained embeddings. The main limitations are the silver (rule-based) labels, keyword-derived topic labels with a very small Economy class, and stemming that over-merges words. Gold annotation and more data would be the most valuable next steps. All code, data splits, models and results are available at https://github.com/SarmadRazaq/i24-7709-NLP-Assignment2.